HYDROPURE
Transforming water

STANDARD OPERATING PROCEDURE

FOR

OPERATION AND MAINTENANCE OF PW S&D SYSTEM

CLIENT NAME	GINNI FILAMENTS LTD.
LOCATION	PANOLI
SYSTEM	PURIFIED WATER STORAGE & DISTRIBUTION SYSTEM
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PWD/21

MANUFACTURE AND SUPPLIER

Hydropure System Pvt. Ltd.
412, Diamond Industrial Estate,
Dahisar East, Mumbai 400 068, INDIA
TeleFax: +91 22 289 78725, 2897 9097
www.hydropure.in

TABLE OF CONTENT

R-0	**27-08-2021**	**FIRST DRAFT**	
REV NO.	**ISSUE DATE**	**DESCRIPTION**	**REMARKS**

1.0 PURPOSE

The purpose of this document is to define the procedure for operation and maintenance of PW Storage and Distribution System.

2.0 SCOPE

This SOP is applicable to PW Storage and Distribution System. Details of system are as below

MAKE	PEAK LOAD	EQUIPMENT CODE	LOCATION
Hydropure Systems Pvt. Ltd.	14000 LPH	___	Purified Water Distribution Room

3.0 REFERENCES, ATTACHMENTS AND ANNEXURE

3.1 REFERENCE

3.1.1 Control Panel Operation

3.1.2 Control Philosophy of PW S&D System

3.1.3 P & ID Diagram

3.2 ATTACHMENT

3.2.1 Attachment – 1 Equipment Log Sheets

3.2.2 Attachment – 2 System Hot water Sanitization Records

3.2.3 Attachment – 3 EVF Cartridge Replacement Record

3.2.4 Attachment – 4 Status Label

3.2.4.1 System Sanitization Status

3.2.4.2 EVF Cartridge Replacement Status

3.3 ANNEXURE

3.3.1 ANNEXURE – 1 - Maintenance Schedule sheet

This document elaborates maintenance schedule of critical instruments, which need regular precautionary measures apart from calibration

4.0 RESPONSIBILITIES

4.1 OPERATOR

4.1.1 To operate the PW Distribution plant as per define procedure

4.1.2 To maintain records in log sheets

4.2 SUPERVISOR

4.2.1 To ensure that operation of PW Distribution plant is carried out as per define procedure

4.2.2 To provide training to operator

4.2.3 To ensure the cleaning, operation & preventive maintenance of PW Distribution plant is carried out as per define procedure

4.2.4 To ensure that proper records are maintain

4.3 DEPARTMENT HEAD

4.3.1 To ensure that proper operation & maintenance is carried as per SOP

4.3.2 To ensure implementation of SOP

4.4 QUALITY ASSURANCE DEPARTMENT

4.4.1 To ensure that the job is carried out as per procedure

4.4.2 To ensure that proper records are maintain

4.5 QUALITY HEAD

4.5.1 To review and approve new or revised SOP's

4.6 FACTORY HEAD

4.6.1 To review and approve new or revised SOP's

5.0 DISTRIBUTION

5.1 Engineering Department

5.2 Quality Assurance Department

6.0 DEFINITION OF TERMS

6.1	ADV	:	Actuated Diaphragm Valve
6.2	AFR	:	Air Filter Regulator
6.3	AV	:	Auto Valve
6.4	Bar	:	Bar is a unit of pressure equal to one million (10^6) dynes, equivalent to 10 newtons, per square centimeter
6.5	BPRV	:	Back Pressure Regulating Valve
6.6	BV	:	Ball Valve
6.7	CF	:	Cartridge Filter
6.8	CPG	:	Compound Pressure Gauge
6.9	CPG	:	Compound Pressure Gauge
6.10	CS	:	Client’s Scope
6.11	CT	:	Conductivity Transmitter
6.12	DV	:	Diaphragm Valve
6.13	EVF	:	Electrical Vent Filter
6.14	FT	:	Flow Transmitter
6.15	GFL	:	Ginni Filaments Ltd
6.16	HMI	:	Human Machine Interface
6.17	HPS	:	Hydropure Systems Pvt. Ltd.
6.18	KL	:	Kilo Litres
6.19	LPH	:	Litres Per Hour
6.20	LT	:	Level Transmitter
6.21	M^3/hr	:	Unit of volume, Cubic Meter Per Hour
6.22	O/P	:	Out Put
6.23	OD	:	Outer Diameter
6.24	P& ID	:	Piping and Instrumentation Diagram
6.25	PG	:	Pressure Gauge
			PH = Log_{10} 1/(H^+) = -Log_{10} (H^+)’
6.26	pH	:	PH is defined as the logarithm of reciprocal of the hydrogen ion concentration or is equal to the logarithm of the hydrogen ion concentration with negative sign
6.27	PLC	:	Programmable Logic Controller

6.28	PLP	:	Process Loop Pump
6.29	PT	:	Pressure Transmitter
6.30	PV	:	Piston Valve
6.31	PWD	:	Purified Water Distribution
6.32	PWS&D	:	Purified Water Storage & Distribution
6.33	QA	:	Quality Assurance
6.34	S&D	:	Storage & Distribution
6.35	SB	:	Spray Ball
6.36	SP	:	Sampling Valve
6.37	SS	:	Stainless Steel
6.38	ST	:	Steam Trap
6.39	STR	:	Strainer
6.40	T	:	PW Storage Tank
6.41	TG	:	Temperature Gauge
6.42	TOC	:	Total Organic Carbon
6.43	TT	:	Temperature Transmitter

7.0 PROCEDURE

7.1 SAFETY PRECAUTIONS

7.1.1 Read 'Operating and Maintenance Manual / Instruction Manual' of individual equipment carefully before proceeding with operation of pw distribution system. Safety precautions specified in these manuals shall be followed carefully. Do not deviate from the data specified in these manuals

7.1.2 Wear hand gloves and face mask while handling chemicals if required

7.1.3 Follow safety processes published by respective EHS department

7.2 MODE OF OPERATIONS

There are two service modes in the system as follows

7.2.1 **AUTO MODE –** It is always recommended to operate the system in '**AUTO MODE**' through PLC control panel

7.2.2 **MAINTENANCE/MANUAL MODE –** Means manual operation of plant. It is advisable to use '**MAINTENANCE MODE**' only during maintenance or incase of failure or replacement of any equipment or component. Proper records of these activities shall be maintained in respective sheet.

These modes are selectable from HMI and at a time only one mode can be executed. User has to touch 'Maintenance Mode' icon on HMI to select the Maintenance mode and same way 'Auto Mode' icon to select auto mode operation. If user has selected the Maintenance mode, then 'Auto Mode' selection icon will be disabled and will become enabled only when 'Maintenance Mode' icon is de-selected and vice a versa.

7.2.3 **SANITIZATION MODE –** Hot water sanitization of 'PW Distribution System' will be selected from control system (HMI). Sequence of complete sanitization cycle will be performed in 'Auto Mode' by PLC control system once this mode is selected.

7.3 POSITION OF VALVES

Following procedures are to be followed during working of the system

7.3.1 Position of Manual Valves – Position of manual valves recommended in below table will remain same for 'Auto' as well as 'Manual' mode of operations

Refer below table for the position of manual valves of system-

Position of manual valves specified in below table refers to service mode (either in AUTO or MANUAL mode) i.e. PW is continuously circulated through S & D system

DO NOT FORGET TO CHANGE position of respective manual valves of system after pw sanitization. OTHERWISE it may cross contaminate the system

SR. NO.	MANUAL VALVE TAG NO.	DESCRIPTION OF VALVE	POSITION OF MANUAL VALVE (DURING SERVICE MODE)
A	**SERVICE MODE OF PW S&D SYSTEM**		
I	**PW STORAGE TANK, T – 301 AND ITS ACCESSORIES**		
1.	BV-301	Air vent for PW Storage Tank jacket	Closed
2.	BV-302	Condensate Valve of PW Storage Tank Jacket, T – 301	Open
3.	BV-303	Condensate Valve of PW Storage Tank Jacket, T – 301	Open
4.	DV-303	AT EVF-301	Closed
5.	BLV-301	Drain Valve of PW Storage Tank, T – 301	Closed
II	**PW DISTRIBUTION SKID: LOOP**		
6.	DV-301	Suction Valve of PW Distribution Pump, PLP – 301/302	Open Completely
7.	DV-302	Discharge Valve of PW Distribution Pump, PLP – 301/302	Open Partially *

SR. NO.	MANUAL VALVE TAG NO.	DESCRIPTION OF VALVE	POSITION OF MANUAL VALVE (DURING SERVICE MODE)
8. .	BLV-314	At Return Line	Open Completely
B	**SANITIZATION MODE OF PW S&D SYSTEM**		
I	**PW STORAGE TANK, T – 301 AND ITS ACCESSORIES**		
9.	BV-301	Air vent for PW Storage Tank jacket	Open Initially #
10.	SV-301	At PW Storage Tank jacket	Open (Setable)
11.	BV-302	Condensate Valve of PW Storage Tank Jacket, T – 301	Open
12.	BV-303	Condensate Valve of PW Storage Tank Jacket, T – 301	Open
13.	DV-303	AT EVF-301	Open
14.	BLV-301	Drain Valve of PW Storage Tank, T – 301	Closed
II	**PW DISTRIBUTION SKID: LOOP**		
15.	DV-301	Suction Valve of PW Distribution Pump, PLP – 301/302	Open Completely
16.	DV-302	Discharge Valve of PW Distribution Pump, PLP – 301/302	Open Partially *
C	**SAMPLING VALVES OF COMPLETE SYSTEM**		
17.	Sampling valves shall be closed throughout the operation (Auto OR Manual) of system. Sampling valves shall only be opened only for collection of sample.		

*** This setting will be done by Supplier's personnel during commissioning and shall not be changed unless and until there is change in feed water quality or downstream demand. Contact supplier before making any changes**

Discharge valve of any pump shall be opened partially before starting system if this setting needs to be changed. Then start the system (Auto Mode or Manual Mode) and adjust this valve to get the desired recommended flow to avoid any damage to downstream equipment

 # Air vent valve is required to open initially to remove air present inside the PW storage tank jacket during sanitization mode

7.3.2 POSITION OF AUTO VALVES –

7.3.2.1 **In 'Auto Mode' -** operation of auto valves will be taken care by PLC control panel

7.3.2.2 **In 'Manual Mode' -** auto valves shall be operated manually through HMI screen. Sequence of this operation shall be listed in 'Operation in Manual Service Mode' section of this document

8.0 SANITIZATION OF PW STORAGE AND DISTRIBUTION SYSTEM

8.1 CRITERIA FOR HOT WATER SANITIZATION:

8.1.1 If any parameter out of microbial OR Endotoxin count, goes beyond maximum allowable limit specified by regulatory guidelines

8.1.2 **OR** as a schedule maintenance activity

8.2 SANITIZATION FREQUENCY:

8.2.1 Once in a **MONTH ± 1 week** (the same can be validated during PQ) or criteria mentioned in point no. 8.1, whichever is earlier

8.3 PREREQUISITES

8.3.1 Ensure that all required utilities are connected and available at respective terminal

8.3.2 Display warning sign board stating 'HOT SURFACE DO NOT TOUCH' near PW Storage & Distribution system.

E.g. 

8.3.3 Inform end users about PW sanitization of system so that they will schedule their work accordingly

8.3.4 STOP PW Storage & Distribution system if it is working in AUTO / MANUAL mode

8.3.5 Then put the system in 'Sanitization Mode' by touching the icon 

8.4 **PROCEDURE:** System will perform complete event as mentioned in the 'Control Philosophy' of 'PW Storage and Distribution System'.

8.5 Drain the entire system as mentioned below after completion of the PW Sanitization

8.5.1 **TANK DRAIN –** PW tank and loop will be filled with hot water. We have to empty the system completely.

8.5.2 Manually open the tank drain valve, BLV – 301 to empty tank, T – 301

8.5.3 **POSITION OF POU VALVES –** Initially open all POU valves to remove water if present in system and then close them till further instructions from water system operator

8.6 Note down all parameters in 'Equipment Log Sheet'

8.7 After completion of the 'SANITIZATION MODE', START the PW storage & distribution system in Auto/Manual Mode

9.0 REPLACEMENT OF VENT FILTER CARTRIDGE (EVF – 301)

9.1 REPLACEMENT FREQUENCY:

9.1.1 Once in a **SIX MONTHS ± 1 week** OR if any physical damage **OR** whichever is earlier (the same can be validated during PQ)

9.2 PROCEDURE:

9.2.1 REPLACEMENT OF FILTER ELEMENT OF EVF:

9.2.1.1 Open the cartridge filter cover through TC clamp which is holding body on top of the tank

9.2.1.2 Remove the existing filter element & replace it with the new filter

9.2.1.3 Record the Make, Model no. and Sr. No. or Part No. of newly installed cartridge in log sheet

9.2.1.4 Observe the inside surface of filter body, clean it with the help of soft clean cloth if any particulates, abnormal carryover of any other material is there inside the filter body

9.2.1.5 If required wash the filter body with purified water

9.2.1.6 Check for any external & internal damages on the body of the filter housing

9.2.1.7 Install new cartridge filter elements and fix the filter housing by tightening TC clamp

Specifications of new cartridge filter shall match with the specification of existing filter cartridge

9.2.1.8 Now EVF is ready to take in service

9.2.1.9 Records this data in respective log sheet

10.0 MAINTENANCE

Refer **ANNEXURE – 1** ‘Maintenance Schedule’ for preventive maintenance of critical instruments.

Every Breakdown of Equipment should be recorded in Equipment History Card.